ZAKR(TM) NV-BAND

Real-Time Timing Budget and Latency Analysis

1. SYSTEM TIMING OVERVIEW

The ZAKR embedded inference and stimulation pipeline is designed for deterministic low-latency operation on the ARM Cortex-M33 architecture. All major execution stages are bounded with explicit timing constraints.

Primary objectives:

- Sub-10 ms AI inference
- <100 ms end-to-end protocol validation
- <300 ms emergency response activation
- Deterministic scheduling under worst-case load

2. STANDARD INFERENCE PIPELINE

Figure B1. Standard AI Inference Timing Pipeline

Stage	Mean Time	Worst Case
EEG acquisition window	4.0 ms	5.0 ms
Feature extraction	2.45 ms	3.1 ms
INT8 LSTM inference	2.02 ms	2.8 ms
Fully connected layer	1.30 ms	1.8 ms
Post-processing	0.86 ms	1.2 ms
Safety checks	0.50 ms	0.9 ms
Logging overhead	0.57 ms	0.8 ms
Total	7.20 ms	9.80 ms

3. FEATURE EXTRACTION TIMING

CMSIS-DSP Operations

Spectral Processing

- FFT length: 256 samples
- Processing time: 0.92 ms

PAC Computation

- Hilbert transform
- Phase extraction
- Amplitude coupling
- Time: 0.74 ms

Sensor Fusion

- PPG normalization
- EDA normalization
- Motion artifact suppression
- Time: 0.79 ms

Total feature extraction: 2.45 ms average.

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4. LSTM INFERENCE PERFORMANCE

Hardware Platform

Processor:

- ARM Cortex-M33
- 64 MHz operating frequency

Optimization:

- CMSIS-NN kernels
- TensorFlow Lite Micro
- INT8 quantization

Inference Benchmark Results

Metric	Value
Mean latency	7.2 ms
Standard deviation	0.4 ms
Minimum	5.9 ms
Maximum	8.9 ms
Target threshold	8.0 ms

1000-cycle simulation demonstrated stable operation within the design target envelope.

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5. FLOAT32 VS INT8 COMPARISON

Execution Mode	Mean Latency
Float32	27.4 ms
INT8 quantized	7.2 ms

Observed speedup: 3.8x.

Benefits:

- Reduced SRAM usage
- Lower power consumption
- Improved thermal stability
- Real-time scheduling margin

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6. SAFETY VALIDATION LATENCY

PCS Validation

Checks:

- Template validity
- Montage validation
- Ramp rate validation
- FFT waveform integrity

Execution time:

- Mean: 1.1 ms
- Worst case: 1.7 ms

SELVL Validation

Checks:

- Current limits
- Frequency limits
- Session duration
- Daily session quota

Execution time:

- Mean: 0.3 ms
- Worst case: 0.5 ms

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7. STIMULATION ENABLE LATENCY

Hardware Activation Chain

Validated protocol -> DAC configuration -> Polyfuse monitoring -> Isolation verification -> Current enable

Timing:

- DAC configuration: 0.8 ms
- Hardware verification: 0.6 ms
- Output stabilization: 1.2 ms

Total: 2.6 ms average.

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8. EMERGENCY PIPELINE TIMING

ZAKR HELP Pathway

Event	Target Time
Wakeword detection	0 ms
Emergency flag raise	50 ms
GNSS polling	100 ms
BLE transmission	150 ms
SMS dispatch	200 ms
Total target latency	<300 ms

No stimulation commands are executed through the emergency pipeline.

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9. CONTINUOUS MONITORING SCHEDULER

Real-Time Task Allocation

Task	Frequency
EEG acquisition	250 Hz
PAC computation	10 Hz
LSTM inference	10 Hz
Safety verification	Every cycle
Audit logging	Every event
BLE synchronization	1 Hz

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10. POWER MANAGEMENT TIMING

Duty Cycling

Inference active: ~65 ms

Sleep/idle window: ~35 ms

Cycle budget: 100 ms total at 10 Hz.

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11. TIMING GUARANTEE CONDITIONS

Guaranteed under:

- Full sensor operation
- BLE active state
- Concurrent logging
- Battery >20%
- Ambient temperature 0-45 C

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12. CONCLUSION

Simulation and benchmark analysis demonstrate that the ZAKR NV-BAND inference and safety pipeline satisfies deterministic low-latency requirements on ARM Cortex-M33 hardware. INT8 quantization enables stable sub-10 ms inference while maintaining sufficient safety verification overhead and scheduling margin.